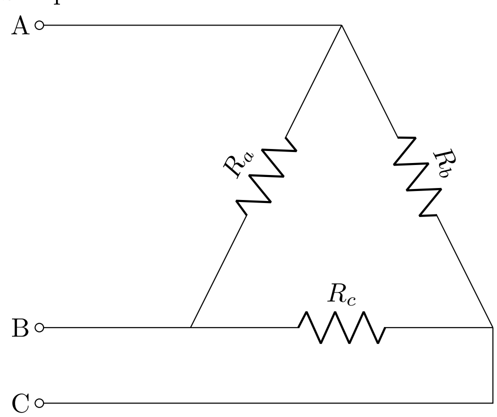
Experiment 2
Circuit Laws and Network Theorems-II

Components/Instruments/Meters

- Breadboard
- Resistances
- DC power supply
- Connecting wires

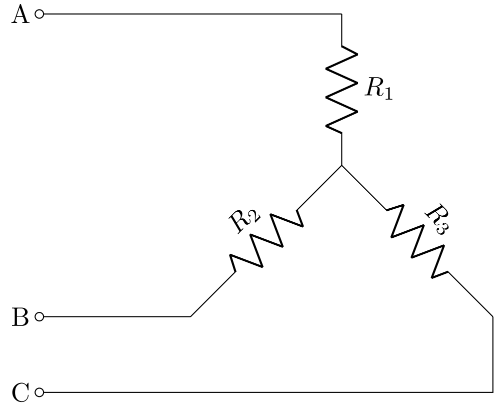
Part 1 (Δ-Y)

1. Setup a Δ circuit as shown below



2. Use $R_a = R_b = R_c = 150\Omega$.
3. Write down the expressions for the effective resistances between the terminals A-B, B-C and A-C in terms of R_a, R_b and R_c.
4. Use your expressions in point 3 above to compute the effective resistances between the terminals A-B, B-C and A-C.
5. Measure the resistances between the terminals A-B, B-C and A-C. Compare your readings with point 4 above.

6. Now connect a Y circuit as shown below



7. Take $R_1 = R_2 = R_3 = 50\Omega$.
8. Once again, write down the expressions for the effective resistances between the terminals A-B, B-C and A-C, now in terms of R_1, R_2 and R_3.
9. Use your expressions in point 8 above to compute the effective resistances between the terminals A-B, B-C and A-C.
10. Measure the resistances between the terminals A-B, B-C and A-C.
11. Compare the measured values in point 10 with the measured values for Δ connection in point 5.
12. Refer to your answers in points 4 and 8 above and equate the effective resistances between the three terminals for Δ and Y connections.
13. Work out the expressions for R_1, R_2 and R_3 in terms of R_a, R_b and R_c. Verify your expression with the values of R_a, R_b and R_c, and R_1, R_2 and R_3 used above.
14. Now use the following values of the resistances R_a, R_b and R_c

R_a	50
R_b	100
R_c	150

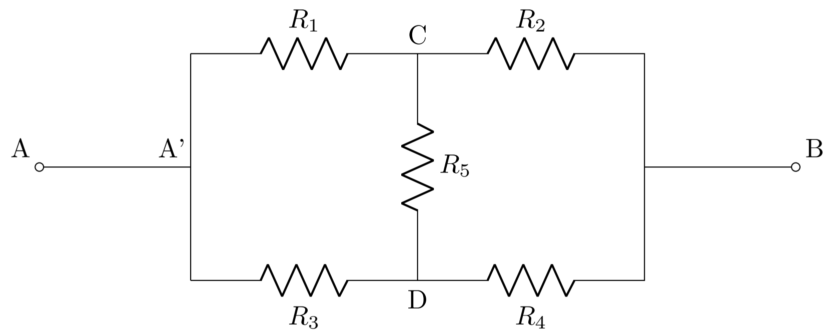
R_1	?
R_2	?
R_3	?

Using your answer to point 13, can you compute the values of R_1, R_2 and R_3 such that the effective resistances between A-B, B-C and B-C for Δ connection are respectively, the same as the effective resistances between A-B, B-C and B-C for Y connection?

15. Verify your results experimentally using the resistances available in lab.

Part 2 (Bridge circuits)

1. Setup a circuit as shown below. Each resistor is 120Ω.



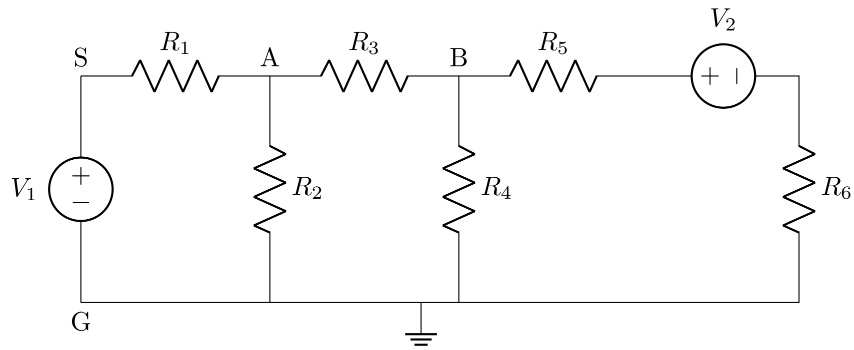
2. Measure the resistance between nodes A and B.
3. What is the expected resistance between nodes A and B? Does the expected value agree with the measured value? (One way to compute the effective resistance between A-B is to simplify the circuit using the Δ-Y transformation learnt in Part 1.)
4. Apply a voltage of 5 volts between A and B.
5. Measure voltages between B-C, B-D, C-D. Make comments about the observed voltages.
6. Replace C-D with a short-circuit. Measure the voltages across the resistances and calculate the current in A-A'.
7. Replace C-D with an open-circuit. Once again, measure the voltages across the resistances and calculate the current in A-A'.
8. Comment on your observations.
9. Repeat the experiment for the following values of resistances

R_1	220Ω
R_2	330 Ω
R_3	100 Ω
R_4	150 Ω
R_5	120 Ω

10. Comment on your observations.

[Optional] Part 3 (Mystery)

1. Setup a network as shown below



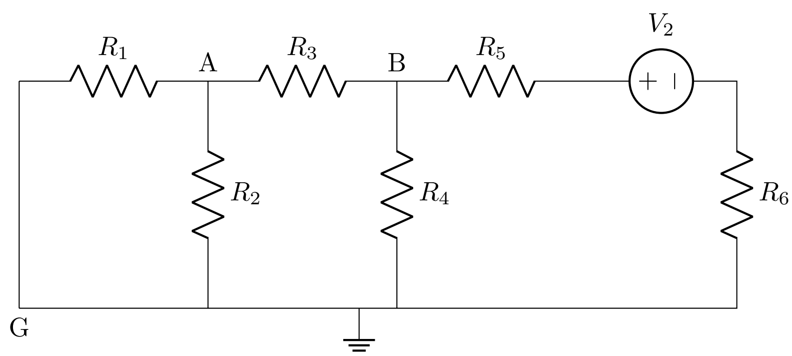
2. Use the following values for the input voltages and resistances

V_1	2V
V_2	3V
R_1	100Ω
R_2	220 Ω
R_3	100 Ω
R_4	220 Ω
R_5	47 Ω
R_6	150 Ω

3. Measure the voltages across R_2 and R_4 and calculate the currents flowing through them.

4. Recall that you had calculated the currents earlier in Experiment 1 when there was only V_1 of 2 volts.

5. Now replace V_1 with a short circuit. Again measure the voltages across R_2 and R_4 and calculate the currents.



6. Now you have 3 sets of readings: (i) with only V_1, (ii) with only V_2, (iii) with both V_1 and V_2. Is there a relation between the three?